

1-C:

A, B and D: Options A, B and D represent “movement along the production possibility curve”. When resources are allocated efficiently, or in other words, an economy is operating at its potential, this means that **“it will not be possible to make someone better off, without making someone else worse off”**

C: Efficiency does not associate between producers. Efficiency in prices cannot be associated with producers. This is why, key D, becomes incorrect.

2-C:

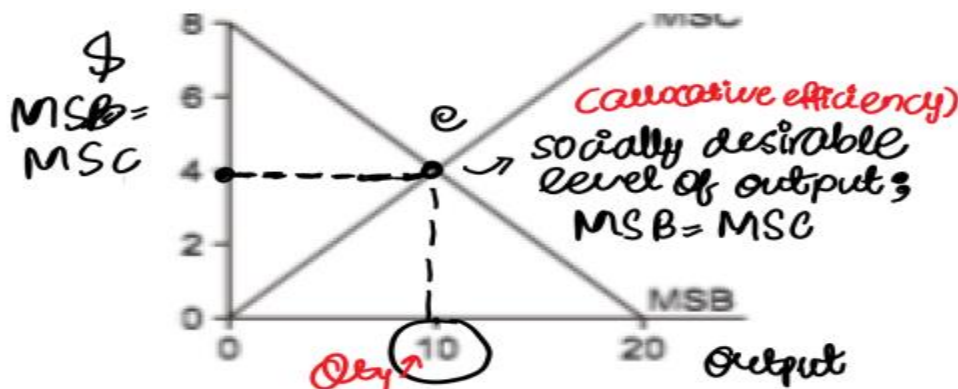
C) A positive externality occurs when an economic activity **generates benefits that spill over to third parties who are not directly involved in the activity**. In the context of the Chinese proverb, **the act of planting a tree by one generation generates a positive externality for the next generation, who can enjoy the shade provided by the tree**. The **benefit of the shade extends beyond the individual or generation that planted the tree**, creating a positive spillover effect for others. This is why, key C, becomes correct.

A) Comparative advantage refers to the ability of a country to produce a good or service at a lower opportunity cost than others, which results in **“specialization”**. The proverb does not illustrate comparative advantage, as it is not about different generations specializing in different activities based on their relative opportunity costs, thus making this incorrect.

B) Division of labor involves **breaking down the production process into specialized tasks to increase efficiency and productivity**. The proverb does not relate to division of labor, thus making this incorrect.

D) A **public good** is a type of good that is **non-excludable and non-rivalrous**, meaning that it is available for everyone to use, as payment by one individual does not restrict others from enjoying the benefits associated with these goods, and one person's use does not diminish its availability to others. It does not specifically address goods or services that exhibit the characteristics of public goods, thus making this incorrect.

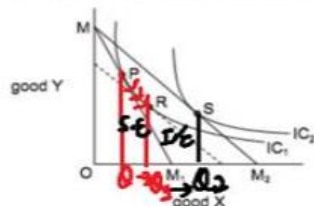
3-C:



- Since, **socially desirable level of output** is at “10 units”, because $MSB = MSC$, option C, becomes correct.
- All other options(A, B and D) are incorrect, because any output below or above this output(10 units) results in under/over allocation, which leads to creation of “dead-weight loss of economic welfare”

4-C:

The diagram shows two indifference curves and two budget lines for two goods X and Y.



The initial position is P. P-R is a substitution effect. R-S is an income effect.

- * Initial budget line: MM_1
- * New budget line: MM_2 (price of X falls)
Budget line pivots rightwards)
- * Substitution effect
(P-R) → quantity demanded extends from Q_1 to Q_1
- * Income effect (R-S) → Qty demanded extends from Q_1 to Q_2 .

A – In case of giffen goods, the income effect outweighs the substitution effect, which is why, this becomes incorrect.

B- We cannot deduce whether this good is a luxury good or not since we do not have information about types of goods

C- This is a normal good because the income effect and the substitution effect move in the same direction. This can be seen in the diagram above.

D – This is incorrect as for the good to be an inferior good, the income effect and the substitution effect must move in the opposite direction.

5-A:

A – When **Average cost** equals **Average Revenue**, this means that the firm is making “normal profits”. This is why, key A, becomes correct.

B – When marginal cost equals Average cost, this represents the point where the marginal cost curve intersects the Average cost curve, which shows the minimum of Average cost.

C – When marginal cost equals Average revenue(price) , this represents “allocative efficiency”

D – When marginal cost equals Marginal revenue, this represents the “profit-maximizing level of output”

6-D:

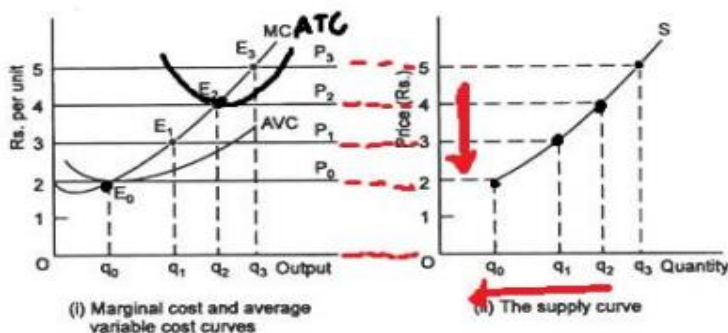


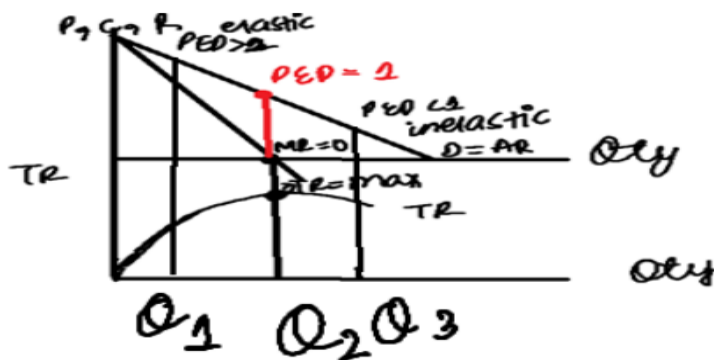
Fig. 7 : The Supply Curve for a Price-taking Firm

In the short-run, the equilibrium is determined where $MC = MR$. This point on the marginal cost curve shows a particular qty that will be supplied at the ruling market price. If the price falls, then the producer will adjust its output such that MC is equal to new MR . This goes on if the price continues to fall until it reaches the minimum of AVC , which is the last operating point in the short-run. **In this way, we get a series of points along the marginal cost curve, which shows a positive relationship between price and quantities a producer is willing to supply.**

Therefore, $MC = \text{Supply}$, and this is how the marginal cost curve helps to derive the market supply curve in a perfectly competitive industry. This is why, key D, becomes correct.

In contrast, where there is an existence of imperfect competition (key A), examples of which include: a monopoly (option B) and an oligopoly (option C), MC is not equal to Supply because of existence of “price-discrimination”. Importantly, there is no such role of supply in this market structure, which makes them all incorrect.

7-B:



- ➔ The diagram above explains that at output Q_1 , the value of PED will be “greater than 1”, and at output of “ Q_2 ”, it will be “=1”, and at output of Q_3 , it will be “less than 1”
- ➔ This is why, key B, becomes correct.

8-A:

→ The “concentration ratio”(CR) in economics compares the sales of a specified number of the “largest firms” in the industry, with the industry’s total sales. In other words, it calculates the total percentage market shares of the specific number of largest firms in the industry.

To calculate the “5-firm concentration ratio”, we are required to add in the market shares of the five largest firms.

To calculate the “3-firm concentration ratio”, we are required to add in the market shares of the three largest firms.

In September 2008 Google introduced its Chrome web browser. The table shows the market percentage (%) share of different web browsers between September 2008 and June 2014.

web browser	% share in September 2008	% share in June 2014
Internet Explorer	67.2	21.0
Firefox	25.8	17.9
Safari	3.0	10.3
Opera	2.8	1.8
Chrome	1.0	45.5
others	0.2	3.5

3 firm concentration ratio:-

$$\times 2008: 67.2 + 25.8 + 3.0$$

$$\Rightarrow 96$$

$$\times 2014: 45.5 + 21.0 + 17.9$$

$$\Rightarrow 84.4$$

It fell

* 5 firm concentration ratio in 2008:

$$\times 67.2 + 25.8 + 3.0 + 2.8 + 1.0$$

$$\Rightarrow 99.8$$

in 2014

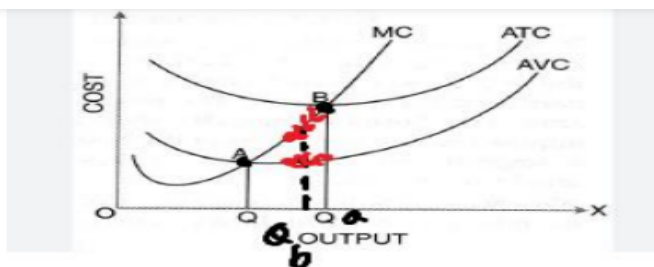
$$\Rightarrow 45.5 + 21.0 + 17.9 + 10.3 + 3.5$$

$$\Rightarrow 98.2$$

It fell.

This explains why, key A, will be correct.

9-A:



Initially, we are operating at quantity Q_a , that is, when minimum, efficient to scale point has been achieved at which long run average cost is minimized

Next, as we reduce the quantity by little from Q_a to Q_b , we can see that both, the average variable cost and marginal cost will decline.

Both of them will decline until they reach quantity "Q", until both: marginal cost, and average variable cost are equal, or in other words, they intersect, to achieve the lowest point of "average variable cost"

10-A:

Between year 1 and year 2, sales value is **rising**. Pre-tax profits are **falling**. Net capital employed is also **falling**. "Net capital employed provides a measure of the company's long-term investment in its operations. It reflects the amount of capital that is available to fund the company's core activities, such as purchasing assets, financing operations, and generating returns for shareholders." But since, **it can be seen that two variables are falling, we cannot say that the firm has definitely grown in size.**

Between year 2 and 3, the sales value is **falling**. And, pre-tax profit is also **falling**. Importantly, net capital employed is rising. But since, **it can be seen that two variables are falling, we cannot say that the firm has definitely grown in size.**

Between years 1 and 3, the sales value is **rising**. And, also net capital employed is **rising**. But, pre-tax profit is **falling**. Since, two out of three variables are rising, it can be said that the **firm has certainly grown in size**

This is why, key A, becomes correct

11-D:

A – Economies of scale and dis-economies of scale are related to average total cost, and not average revenue.

B – Management policy changes is related to diseconomies of scale, arising as a result of integration, for example, horizontal integration, or because of conglomerate growth (diversification)

C – Economies of scale and diseconomies of scale occur in the long-run instead of short-run.

D – Economies of scale occur with "decreasing Average total cost" arising from a firm growing in size , as compared to dis-economies of scale that occur because of "increasing Average total cost".

12-D:

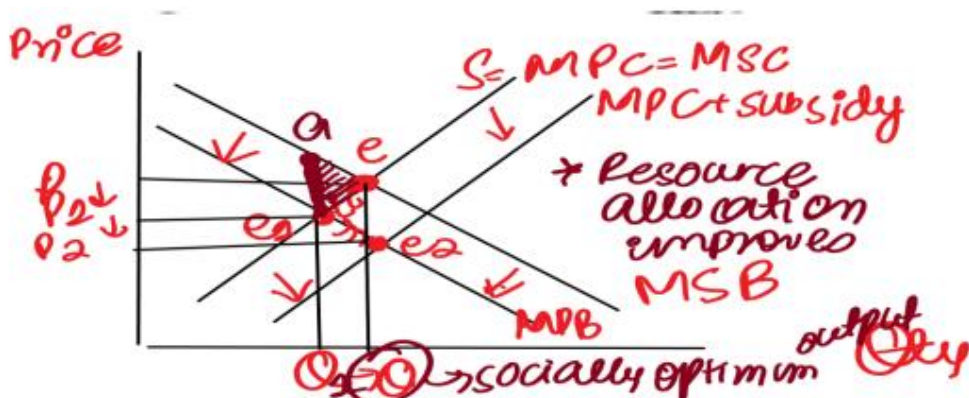
A – When a firm finances new investments mainly from retained profits, this represents that the objective of "profit-maximization" will be there to facilitate it.

B – When firms have active shareholders, this means that the objective of firms is mostly likely going to be "profit-maximization" since this will lead to them getting higher dividend pay-outs.

C – When a company buys another company with the financial assistance of investment banks, this means that, for arrangements of finances, further shares will not have to be issued, which will prevent the original shareholders from losing control, or in other words, it will prevent the dilution of ownership. Now this means that the objective is most likely to be "profit-maximization"

D - When there are a large number of shareholders , which represents a large public ltd company, and employed paid managers, this will give existence to the principal-agent problem, that occurs as a result of "divergence of interests". Divorce of ownership and control will exist because managers who are in control may want to achieve the objective of sales-revenue maximization since their salaries and bonuses are based on revenues, and not profits, as compared to shareholders, who aim for "profit-maximization", to be able to receive a higher payout in form of dividends. This is why, key D, is correct.

13-A:



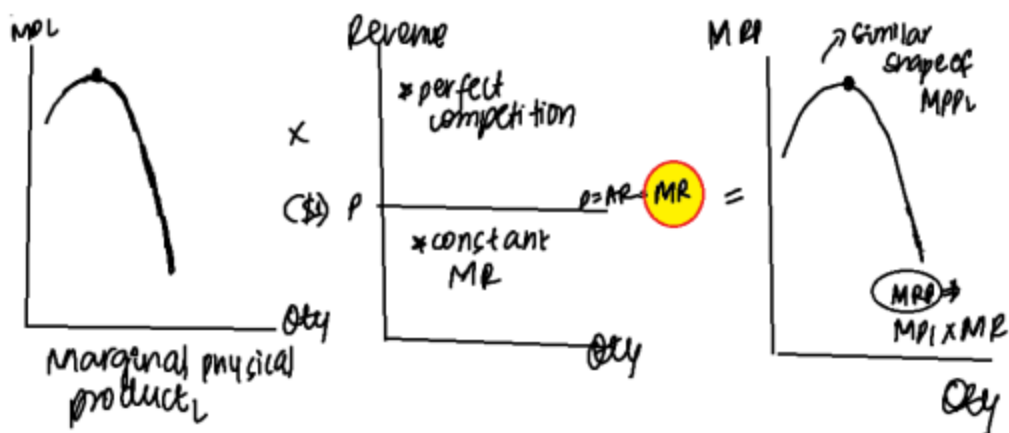
A – Initially, when qty of “Q1” was produced, there was under-allocation of resources, which resulted in deadweight loss of “ $ae1$ ”. But, now after subsidy, when supply increases from MSC to MPC+subsidy, the output of “Q”, which is the socially desirable level of output is produced, which means that **resource allocation will improve**. This is why, key A, will be correct.

B – The firm will not produce less, instead output is shown to rise from Q1 to Q.

C – There will not be a misallocation of resources; instead the allocation of resources improves.

D – This is incorrect as there will be an effect on production, because output will rise from Q1 to Q.

14-B:



A and C: Options A and C are incorrect because in case of a “perfectly competitive market structure”, the Average revenue and Marginal Revenue are same.

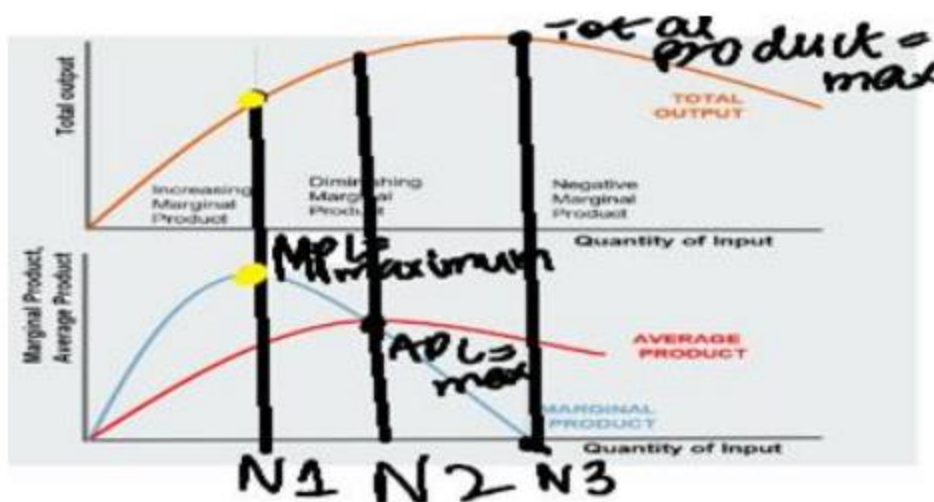
B : Marginal revenue productivity represents the marginal revenue which the firm gets from employing one more worker, and it is the value of the “Marginal product of labor”, obtained by multiplying “Marginal physical product of labor” by “Marginal Revenue”, because it is assumed that the physical productivity of labor can be measured.

Marginal Revenue is the revenue received by selling an additional unit, and it is constant under “perfect competition”, and the “marginal product of labor curve” follows the “law of diminishing returns” i.e. initially Marginal product of labor curve increases due to “increasing returns”, and then it falls due to “decreasing returns”. Therefore, the Marginal revenue productivity curve is the shape of the Marginal product of labor curve because it is the product of Marginal Product of labor curve and a constant Marginal Revenue. **So, when the “Marginal physical product of labor” falls, this would in turn lead to the “Marginal Revenue productivity” to fall too. This explains why key B is correct.**

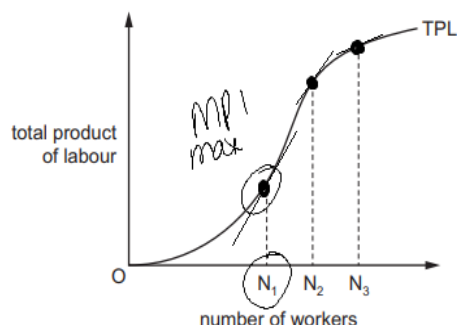
D – The supply price of the factor does not rise because the “supply of labor is perfectly elastic” in a perfectly competitive labor market.

15-B:

Let us first understand the concept of **MAT theory**. This means that first, marginal product of labor reaches its maximum, followed by Average product of labor and total product of labor.



Now, I will be telling the methodology to deduce the level of workers at which marginal product of labor reaches its maximum.



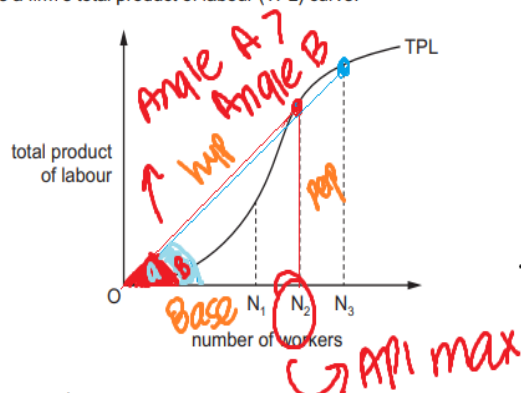
The slope of TPL curve is MPL.

We will first make a line tangent to each level of workers to find out the slope of the curve at each output level.

The number of workers at which the slope is the steepest is where Marginal product of labor will be at its maximum, which in this case is ON1 number of workers.

Next, I will be telling the methodology to deduce the level of workers where the Average product of labor reaches its maximum.

The diagram shows a firm's total product of labour (TPL) curve.



✦ We will make 90 degree angles. The first coordinate will be the origin. The second will be the point on the TPL curve corresponding to the respective level of workers. And the third point will be at the respective level of workers. The one which has the largest angle between the hypotenuse and the base denotes the level of workers at which the APL will be at its maximum.

✦ Therefore, APL is maximised at ON_2 workers

Next, we are already aware that TPL will be at its maximum at a level beyond ON_3 number of workers.

A- The Average product of labor is maximized at “ ON_2 ”, that is, when the marginal product of labor curve intersects the average product of labor curve. This can be seen in the figure above.

B- This is correct that the average product of labor curve is maximized at “ ON_2 ”. This can be seen in the figure above.

C- The marginal product of labor is maximized at “ ON_1 ”, beyond which, the law of diminishing returns sets in

D- At ON_3 , the marginal product of labor is nearly zero, because Total Product of labor is about to reach its maximum.

16-B:

A – A higher rate of sales tax will have an effect on the “product market”

B – A rise in the price of a final product, as a result of perhaps, a rise in the demand for the product represents that now the demand for labor (which is a derived demand), which is also called the “Marginal Revenue product of labor” increases, or in other words, shifts to the right.

Also, $MRP = MPP \times MR$ (which is equal to AR (price) in case of perfect competition), so when MR increases, automatically the MRP will rise. This is why, none other than key B, can be correct.

C – An increase in labor supply will cause the supply of labor or the Average cost of labor curve to shift rightwards.

D – An increase in wages will cause a movement along the demand for labor or the Marginal revenue productivity curve.

17-D:

The argument described is known as the Laffer Curve, which suggests that as income tax rates increase, tax revenue initially increases, but eventually decreases as higher tax rates discourage economic activity and reduce taxable income. This implies that there are limits to the redistributive effects of progressive taxation, as higher tax rates may not necessarily lead to greater tax revenue or reduced income inequality.

This is because, beyond a certain optimal level of tax rate, individuals choose to **evade tax, or in other words, they find loopholes to avoid tax. This explains why key D will be correct.**

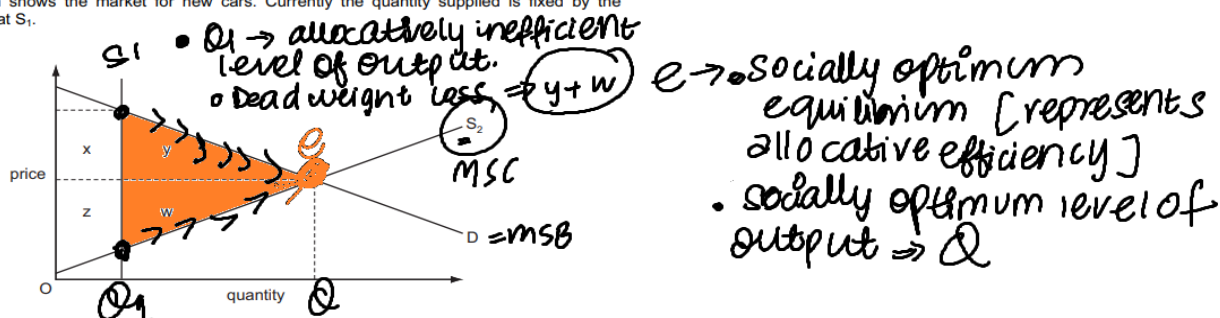
A) The argument does not suggest that governments should forgo attempts to relieve poverty via fiscal policy. Rather, it highlights the limitations of using progressive taxation as a tool for redistribution.

B) While maximizing government tax revenue might be a goal, it is not the primary implication of the Laffer Curve. Rather, it highlights the limitations of using progressive taxation as a tool for redistribution.

C) The argument does not suggest that taxes should be levied on expenditure rather than income.

18-A:

The diagram shows the market for new cars. Currently the quantity supplied is fixed by the government at S_1 .



If the market was deregulated the supply curve would be S_2 .

Which area measures the net welfare gain of removing the restriction on the quantity supplied?

- ☒ A $w + y$ ☐ B $x + y$ ☐ C $z + w$ ☐ D $z + x + y + w$

This explains that when the output increases from Q_1 to Q as a result of removal of the restriction, the dead-weight loss of economic welfare of $y + w$ will be eliminated. This will subsequently also denote the region of net welfare gain.

19-C:

Real GNP (Gross National Product) per head is a measure of the total output of a country which is adjusted for inflation, divided by its population. However, it has its limitations when comparing standards of living across countries. One major issue is the existence of the hidden economy, also known as the informal or shadow economy, which includes unreported income and activities.

Adjusting for the size of the hidden economy in different countries can make real GNP per head a more reliable indicator of standards of living, as it would account for the unreported income and activities that are not captured in official GNP figures. **Since this will subsequently prevent GDP from being understated, this explains why key C will be correct.**

A and D: Exports and government spending will already be adjusted for in real GNP per head figures.

B: Adjusting for inflation rates would already be accounted for in the "real" GNP figure, which is adjusted for inflation.

20-B:

The table shows real GDP per head measured in \$US for countries X and Y over a three year period.

country	2011	2012	2013
X	100	110	121
Y	200	210	231

% change formula:

$$\frac{\text{New} - \text{old}}{\text{old}} \times 100$$

A – Each country did not have the same rate of economic growth between 2011 and 2012. This is because, in between 2011 and 2012, real GDP per head rose by 10% in country X, whereas in country Y, it rose by 5% in between 2011 and 2012.

B – This is correct because growth in RGDP/head is higher in country X (of 10%) than in country Y (of 5%) between 2011 and 2012.

C and D: We cannot deduce if population growth was higher in country X than country Y between 2012 and 2013. Also, we cannot deduce if average price level was higher in country X than in country Y throughout the period based on the given information.

21-D:

- Labour-intensive production: Due to limited capital as a result of a lack of **technical progress**, production processes tend to be more labour-intensive.
- Low rate of saving: Developing countries often have lower rates of saving due to limited income, high poverty levels, and limited access to financial services.

A) Low birth rate and dominant primary sector: Developing countries often have higher birth rates and a larger primary sector (agriculture, mining, etc.).

B) Capital-intensive production and rapidly growing population: Developing countries often lack the capital and technology for capital-intensive production, and their populations are often growing rapidly.

C) External debt and low death rate: While developing countries may have external debt, they often also have higher death rates due to limited access to healthcare and other factors.

22-D:

A) The average age of the total population has not necessarily increased, as the percentage of the population in the 65+ age group has decreased.

B) Based on this data, we cannot calculate the death rate in 2012 and 2014. This means we cannot actually see if death rate rose in between 2012 and 2014.

C) The infant mortality rate has not necessarily risen, as the percentage of the population in the 0-15 age group has increased.

D) The dependent population includes the people who are 0-15 years old and those who are 65 years and over.

- The percentage of the population in the 0-15 age group (dependent population) has increased by **5%** from 25% to 30%.
- The percentage of the population in the 65 years and over age group (dependent population) has decreased by **3%** from 15% to 12%.
- $\rightarrow +5\% - 3\% \Rightarrow +2\%$ (**This represents that the percentage of dependent population has increased overall**)

23-C:

Seasonal unemployment occurs when there is a regular pattern of unemployment during certain times of the year, due to changes in consumer demand arising from changes in season. For example:

In Winter, there is a fall in demand faces by industries like construction and tourism, as a result of which, there is a subsequent fall in demand for labor in these industries which is a derived demand. This explains the existence of seasonal unemployment. **This explains why key C is correct.**

A) A general decrease in demand for goods and services would lead to cyclical unemployment, not seasonal unemployment.

B) A lack of necessary skills in the workforce represents occupational immobility which would exacerbate structural unemployment.

D) An unwillingness of workers to move to other parts of the country where there is work represents geographical immobility which would also lead to workers remaining structurally unemployed for a longer period of time.

24-D:

Rising levels of unemployment can have the following combination of cost and benefit:

Cost: An irretrievable loss of output (the economy is not producing at its potential level, resulting in a loss of goods and services)

Benefit: A reduction in inflationary pressure (high unemployment can lead to lower demand for goods and services, and consequently a fall in consumption expenditure results in a decrease in Aggregate demand which can help reduce demand-pull inflation). Therefore, the correct answer is D.

A) A deterioration in human capital is a cost because prolonged periods of unemployment can erode human capital (**hysteresis effect**) but an increase in capital expenditure is not a benefit of rising unemployment. In fact, firms may lower their investment expenditure on capital goods in event of unemployment.

B) An increase in import expenditure is not a benefit of rising unemployment. This is because, when people are unemployed and have less incomes, their consumption of imported goods decreases, which eventually results in decreased import expenditures.

However, a decrease in government tax revenue is a cost.

C) An increase in the rate of economic growth is not a cost of unemployment. This is because, when there is unemployment, consumption expenditure decreases due to less incomes being earned, which subsequently results in a fall in Aggregate demand, consequently leading to a decrease in actual economic growth.

However, it is mentioned that high unemployment can lead to a more responsive workforce. **This is because, workers may become more flexible and willing to accept jobs that are available, leading to a more responsive workforce. This is because they may be more desperate to find employment and therefore more willing to compromise on wages, working conditions, and other aspects of the job.**

Note:- High unemployment can increase the competition for jobs, making workers more eager to adapt and respond to changing labor market conditions.

25-B:

The correct answer is B: an extension of bank opening hours.

An extension of bank opening hours would make it easier for people to access their money from banks whenever needed, reducing the need to hold cash. This would lead to a decrease in the public's desired ratio of cash to bank deposits.

A) A decrease in street crimes would make individuals less likely to hold cash. This will subsequently bring about an increase in public's desired ratio of cash to bank deposits

C) The introduction of a charge per transaction on accounts would likely increase the public's desire to hold cash, as people might try to avoid transaction fees. This will subsequently bring about an increase in public's desired ratio of cash to bank deposits

D) The introduction of annual charges for credit cards might lead to a decrease in credit card usage. Since individuals are no longer purchasing on credit, this implies that there will be an increase in cash purchases. For this, they will need to hold cash, which will again bring about an increase in public's desired ratio of cash to bank deposits

26-C:

Accelerator:

- The investment is induced, meaning it is dependent on changes in real national income.
- The accelerator effect refers to the phenomenon where an increase in the rate with which real national income is rising leads to an increase in induced investment, as firms need to expand their capacity to meet the higher demand.

Multiplier:

- The investment is autonomous, meaning it is independent of the level of income.
- An increase in autonomous investment will subsequently bring about a multiplier times increase in real national income.

27-D:

A) When a change in real national income brings about a change in investment expenditure, this represents the accelerator effect. Therefore, the formula (change in Investment/Change in income) helps to calculate the co-efficient of the accelerator.

B) When a change in investment brings about a change in consumption, this does not help to calculate the value of investment multiplier.

C) When a change in income brings about a change in consumption (change in consumption/change in income), this helps to calculate the marginal propensity to consume.

D) When a change in autonomous investment brings about a change in real-national income, this represents the multiplier effect in action. Therefore, the formula of investment multiplier will be: $\text{Change in Income} / \text{Change in Investment}$. **Let's say investment rose by 100m and the resultant rise in real national income was 500m, this implies that the value of investment multiplier is 5. (500/100)**

28-D:

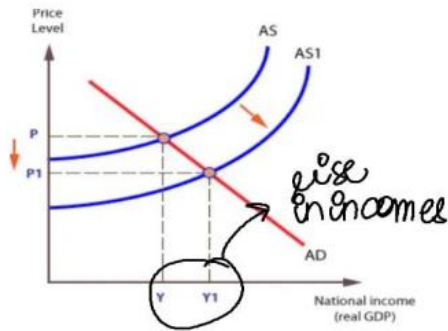
Stabilization policies are aimed to smooth out fluctuations in business cycle and national output, thus making option D correct. This can be seen in the diagram below:



29-D:

If the government of Lesotho provides subsidies to local entrepreneurs, this will have the **effect of increasing the government expenditure**. Moreover, since provision of subsidies is a supply-side policy, it will reduce the costs, and increase Aggregate Supply, which will have the effect of an **increase in incomes from Y to Y1**, as can be seen in the diagram below.

However, although this makes exported goods competitive, whether or not the Balance of trade position improves, depends upon the price elasticity of demand. And also, there are other components of current account as well. This means that the current account position will be uncertain.



* X prices $\downarrow$	* X prices $\downarrow$
X Qty $\uparrow$ elastic demand	X Qty $\uparrow$ inelastic demand
$X(CTR) \uparrow$	$X(CTR) \downarrow$
net position improves	net position worsens

30-D:

Option A: The goal is not to bring potential output up to the level of actual output. While supply-side policies aim to increase potential by raising LRAS, their aim is never to reduce potential output so that it is equal to the level of actual output.

Option B: Ensuring a budget surplus is done through a contractionary fiscal policy, which is separate from supply-side policies.

Option C: Supply side policies never aim to ensure that the composition of output matches the pattern of demand

Option D: Supply-side policies aim to improve the efficiency of the economy by increasing productivity through investments in education and training for instance which indicate an interventionist supply-side tool. This will subsequently result in an increase in Long-run Aggregate supply, which will lead to a rise in the potential output of the economy.

